

# Tables, Modules, Services & APIs

Extracted from the backend design · for the developers building it

|          |                                                     |
|----------|-----------------------------------------------------|
| Document | Reference extract · v1.1                            |
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| Contents | 44 tables · 14 modules · 27 services · 91 endpoints |

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# 1. Database tables

PostgreSQL 16 with TimescaleDB. Every business table carries `tenant_id` and is protected by row-level security; primary keys are UUIDs, and the codes users see (MTR-1001, GW-01, AR-7F2 ...) are separate unique columns. All tables also have `created_at` / `updated_at`.

| Area                             | Tables                                                                                                                                                                                                                                                                                                                 |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Platform and tenants             | <code>platform_admins</code> , <code>tenants</code> , <code>tenant_status_history</code> , <code>tenant_limits</code>                                                                                                                                                                                                  |
| Identity and access              | <code>users</code> , <code>user_sites</code> , <code>invites</code> , <code>auth_sessions</code> , <code>otp_codes</code> , <code>roles</code> , <code>permissions</code> , <code>role_permissions</code> , <code>field_permissions</code>                                                                             |
| Sites, devices and meters        | <code>sites</code> , <code>device_models</code> , <code>meter_models</code> , <code>devices</code> , <code>device_credentials</code> , <code>register_maps</code> , <code>cost_centres</code> , <code>meters</code> , <code>energy_targets</code> , <code>virtual_meter_members</code> , <code>connection_flows</code> |
| Telemetry (time-series)          | <code>readings</code>                                                                                                                                                                                                                                                                                                  |
| Tariffs, shifts and schedules    | <code>tariffs</code> , <code>tariff_instances</code> , <code>shifts</code> , <code>shift_schedules</code> , <code>shift_assignments</code> , <code>shift_instances</code>                                                                                                                                              |
| Alerts and notifications         | <code>alert_rules</code> , <code>alert_rule_state</code> , <code>alerts</code> , <code>alert_events</code> , <code>notifications</code> , <code>notification_preferences</code>                                                                                                                                        |
| Audit, exports and device events | <code>audit_log</code> , <code>export_jobs</code> , <code>device_messages</code> , <code>device_commands</code> , <code>publish_destinations</code> , <code>publish_attempts</code> , <code>device_events</code>                                                                                                       |

## 1.1 Platform and tenants

**platform\_admins** Control Center accounts — a separate realm from tenant users

| Column                                 | Type             | Notes                        |
|----------------------------------------|------------------|------------------------------|
| <code>id</code>                        | uuid PK          |                              |
| <code>username</code>                  | citext unique    | e.g. superadmin              |
| <code>name</code> , <code>email</code> | text             |                              |
| <code>password_hash</code>             | text             | argon2id                     |
| <code>mfa_secret</code>                | text null        | TOTP; required in production |
| <code>last_login_at</code>             | timestamptz null |                              |

**tenants** One organisation (the Org ID users type at sign-in)

| Column                                                                                  | Type                   | Notes                                                            |
|-----------------------------------------------------------------------------------------|------------------------|------------------------------------------------------------------|
| <code>id</code>                                                                         | varchar(12) PK         | Org ID: chosen slug (FEBINO) or generated 8-char code (HX4R7TQM) |
| <code>name</code>                                                                       | text                   | legal name, Febino Solutions Pvt Ltd                             |
| <code>slug</code>                                                                       | citext unique          | Workspace URL, febino; <code>^[a-z0-9][a-z0-9-]{1,29}\$</code>   |
| <code>industry</code>                                                                   | text                   | Manufacturing plant, Data centre, Solar / renewable plant ...    |
| <code>gst_in</code>                                                                     | varchar(15) null       | hidden from some roles by field permissions                      |
| <code>country</code> , <code>timezone</code> , <code>address</code> , <code>city</code> | text                   | timezone default Asia/Kolkata                                    |
| <code>discom</code>                                                                     | text                   | utility, e.g. BESCOM, TANGEDCO                                   |
| <code>contract_demand_kva</code>                                                        | integer                | for maximum-demand checks                                        |
| <code>meter_range</code>                                                                | text                   | 1-10   11-50   51-200   200+ (sets default meter limit)          |
| <code>plan</code>                                                                       | enum                   | starter   growth   enterprise                                    |
| <code>billing_cycle</code>                                                              | enum                   | monthly   yearly                                                 |
| <code>contact_email</code> , <code>phone</code>                                         | text                   |                                                                  |
| <code>status</code>                                                                     | enum                   | pending   active   suspended                                     |
| <code>status_reason</code>                                                              | text null              | e.g. Subscription invoice overdue                                |
| <code>suspended_until</code>                                                            | timestamptz null       | suspension lifts itself at this time (scheduler)                 |
| <code>approved_at</code> , <code>approved_by</code>                                     | timestamptz, uuid null | platform admin who approved                                      |
| <code>currency</code>                                                                   | char(3)                | INR   USD   AED                                                  |
| <code>energy_unit</code>                                                                | enum                   | kWh   MWh (display only; storage is always kWh)                  |
| <code>co2_factor</code>                                                                 | numeric(5,3)           | kg CO2 per kWh, default 0.820                                    |

| Column         | Type            | Notes                                                |
|----------------|-----------------|------------------------------------------------------|
| default_tariff | numeric(8,2)    | ₹/kWh used when no ToD slot matches (UI demo uses 8) |
| deleted_at     | timestampz null | soft delete; data purged after 30 days               |

### tenant\_status\_history Every approve / reject / suspend / reactivate

| Column                 | Type                    | Notes                     |
|------------------------|-------------------------|---------------------------|
| id                     | uuid PK                 |                           |
| tenant_id              | FK tenants              |                           |
| from_status, to_status | enum                    |                           |
| reason                 | text null               |                           |
| until                  | timestampz null         | for temporary suspension  |
| by_admin_id            | FK platform_admins null | null = system (auto-lift) |
| at                     | timestampz              |                           |

### tenant\_limits Overrides of the plan's default limits (Control Center → Limits)

| Column         | Type             | Notes                                                                                            |
|----------------|------------------|--------------------------------------------------------------------------------------------------|
| tenant_id      | FK, PK part      |                                                                                                  |
| key            | enum, PK part    | users   meters   subMeters   devices   sites   tariffs   shifts   schedules   alertRules   roles |
| value          | integer          | 0 = unlimited                                                                                    |
| reason         | text null        | why the admin changed it                                                                         |
| set_by, set_at | uuid, timestampz |                                                                                                  |

## 1.2 Identity and access

### users A person's account inside one tenant (the same email may exist in several tenants)

| Column                  | Type             | Notes                                                                          |
|-------------------------|------------------|--------------------------------------------------------------------------------|
| id                      | uuid PK          |                                                                                |
| tenant_id               | FK tenants       |                                                                                |
| emp_id                  | varchar(16)      | EMP-014; unique per tenant, auto-next                                          |
| first_name, last_name   | text             |                                                                                |
| email                   | citext           | unique per tenant                                                              |
| username                | citext unique    | unique across the platform (sign-in by username works without guessing tenant) |
| mobile                  | varchar(20) null | for OTP sign-in and SMS alerts                                                 |
| role_id                 | FK roles         |                                                                                |
| shift_id                | FK shifts null   | Shift A / B / C / General                                                      |
| department, designation | text null        |                                                                                |
| status                  | enum             | invited   active   inactive   suspended                                        |
| two_factor              | boolean          | require OTP after password                                                     |
| language                | varchar(8)       | en-IN   hi   ta (My profile)                                                   |
| timezone                | text null        | IANA name; null = tenant's                                                     |
| password_hash           | text null        | null until the invite is accepted                                              |
| last_active_at          | timestampz null  | shown as "12 min ago"                                                          |
| invited_by              | FK users null    |                                                                                |

**user\_sites** Optional: restrict a user to some sites

| Column  | Type        | Notes               |
|---------|-------------|---------------------|
| user_id | FK, PK part |                     |
| site_id | FK, PK part | no rows = all sites |

**invites** Invitation links (accept-invite.html?token=...)

| Column      | Type            | Notes                                                              |
|-------------|-----------------|--------------------------------------------------------------------|
| id          | uuid PK         |                                                                    |
| user_id     | FK users        |                                                                    |
| token_hash  | char(64) unique | SHA-256 of the random token; the token itself is only in the email |
| invited_by  | FK users        |                                                                    |
| expires_at  | timestampz      | 7 days                                                             |
| accepted_at | timestampz null |                                                                    |

**auth\_sessions** Signed-in devices (Settings → Your active sessions)

| Column                               | Type       | Notes                                                             |
|--------------------------------------|------------|-------------------------------------------------------------------|
| id                                   | uuid PK    |                                                                   |
| realm                                | enum       | tenant   platform                                                 |
| user_id / admin_id                   | uuid       | one of them                                                       |
| tenant_id                            | FK null    | current tenant (changes on switch-tenant)                         |
| refresh_token_hash                   | char(64)   | rotated on every refresh; reuse of an old one revokes the session |
| remember                             | boolean    | true: 30 days, false: 12 hours (platform: 8 hours)                |
| ip, user_agent, device_label         | text       | "Chrome on Windows · Bengaluru"                                   |
| last_used_at, expires_at, revoked_at | timestampz |                                                                   |

**otp\_codes** One-time codes (verify email, OTP sign-in, password reset, login approval)

| Column              | Type            | Notes                                                      |
|---------------------|-----------------|------------------------------------------------------------|
| id                  | uuid PK         |                                                            |
| purpose             | enum            | verify_email   login_otp   reset_password   login_approval |
| target              | text            | email or mobile                                            |
| user_id / tenant_id | uuid null       |                                                            |
| code_hash           | text            | 6 digits, hashed                                           |
| expires_at          | timestampz      | 10 minutes                                                 |
| attempts            | smallint        | locked after 5 wrong tries                                 |
| consumed_at         | timestampz null |                                                            |

**roles** Built-in and custom roles

| Column       | Type       | Notes                                                                                        |
|--------------|------------|----------------------------------------------------------------------------------------------|
| id           | uuid PK    |                                                                                              |
| tenant_id    | FK tenants |                                                                                              |
| name         | text       | Tenant Admin, Energy Manager, Supervisor, Operator, Technician, Viewer, Auditor, or custom   |
| description  | text       |                                                                                              |
| is_system    | boolean    | Type: Built-in   Custom; built-in roles cannot be deleted; Tenant Admin cannot be restricted |
| landing_page | text       | Default landing page: dashboard   meters   analytics   alerts                                |

**permissions** Catalogue of permission keys (Roles → Permissions tab)

| Column      | Type           | Notes                                                                                      |
|-------------|----------------|--------------------------------------------------------------------------------------------|
| key         | varchar(48) PK | module.action, e.g. meters.export, alerts.acknowledge                                      |
| module      | enum           | dashboard, meters, devices, connections, analytics, alerts, shifts, users, roles, settings |
| action      | text           | view, create, edit, delete, export, approve, acknowledge ...                               |
| description | text           | What does this allow?                                                                      |
| tenant_id   | FK null        | null = platform-defined; tenants may add custom keys                                       |

**role\_permissions** Which keys a role has (the Permission matrix is a view over this)

| Column         | Type                    | Notes |
|----------------|-------------------------|-------|
| role_id        | FK, PK part             |       |
| permission_key | FK permissions, PK part |       |

**field\_permissions** Field-level visibility per role (field-perms.js)

| Column             | Type          | Notes                                                       |
|--------------------|---------------|-------------------------------------------------------------|
| tenant_id, role_id | FK, PK part   |                                                             |
| module, field      | text, PK part | e.g. meters.cost, devices.mac, users.mobile, settings.gstin |
| can_view, can_edit | boolean       | edit implies view; no row = full access                     |

## 1.3 Sites, devices and meters

**sites** A location with its own utility connection

| Column               | Type              | Notes                            |
|----------------------|-------------------|----------------------------------|
| id                   | uuid PK           |                                  |
| tenant_id            | FK                |                                  |
| name                 | text              | Plant 1 · Bengaluru              |
| address              | text              |                                  |
| utility, consumer_no | text              | DISCOM and service number        |
| contract_demand_kva  | integer           |                                  |
| default_tariff       | enum              | tod   flat                       |
| flat_rate            | numeric(8,2) null | ₹/kWh when default_tariff = flat |
| timezone             | text null         | falls back to the tenant's       |
| status               | enum              | active   inactive                |

**device\_models** Hardware catalogue — real gateways / loggers (hw-catalog.js)

| Column                                         | Type         | Notes                                                                                                                |
|------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------|
| id                                             | uuid PK      |                                                                                                                      |
| name                                           | text unique  | Moxa MGate MB3170, Schneider Link150, Schneider Com'X 510, Siemens SIMATIC IOT2050, Teltonika TRB245, Milesight UG65 |
| maker, kind                                    | text         | Moxa · Modbus TCP ↔ RTU gateway                                                                                      |
| ethernet, serial_ports, cellular, wireless, io | text null    | as published by the manufacturer                                                                                     |
| protocols                                      | text[]       | Modbus TCP, Modbus RTU, Modbus ASCII, LoRaWAN                                                                        |
| uplinks                                        | text[]       | modbus_tcp_poll   mqtt_push   https_push                                                                             |
| default_port                                   | integer null | 502, 443, 8883                                                                                                       |
| power                                          | text         | 12–48 VDC                                                                                                            |
| notes, field_node                              | text null    | e.g. Milesight UC501 node per meter                                                                                  |
| is_global                                      | boolean      | shared library; tenants may add their own                                                                            |

**meter\_models** Hardware catalogue — real energy meters (hw-catalog.js)

| Column          | Type                  | Notes                                                                                                              |
|-----------------|-----------------------|--------------------------------------------------------------------------------------------------------------------|
| id              | uuid PK               |                                                                                                                    |
| name            | text unique           | Schneider PM5560, PM2220, EM6400NG, iEM3255, Secure Elite 440, Fronius Smart Meter TS 65A-3, Deep Sea DSE7320 MKII |
| maker, kind     | text                  | Schneider Electric · Power quality meter (PowerLogic PM5000)                                                       |
| meter_class     | enum                  | basic   multi   pq   dc   dg — default class for meters of this model                                              |
| accuracy        | text                  | Class 0.2S (IEC 62053-22)                                                                                          |
| comms           | text                  | 2 × Ethernet (Modbus TCP) · 1 × RS-485 (Modbus RTU)                                                                |
| wiring          | text[]                | 3P4W, 3P3W, 1P2W — the form only offers these                                                                      |
| inputs, extras  | text null             | CT 1 A / 5 A secondary · harmonics to the 63rd                                                                     |
| register_map_id | FK register_maps null | null = load the manufacturer's register list first                                                                 |
| is_global       | boolean               |                                                                                                                    |

**devices** Gateways and data loggers (Device Management)

| Column                       | Type               | Notes                                                                         |
|------------------------------|--------------------|-------------------------------------------------------------------------------|
| id                           | uuid PK            |                                                                               |
| tenant_id, site_id           | FK                 |                                                                               |
| code                         | varchar(16)        | GW-01; unique per tenant                                                      |
| name                         | text               | Main LT Panel Gateway                                                         |
| type                         | enum               | modbus_tcp_gateway   energy_server   edge_nodered   edge_4g   lorawan_gateway |
| device_model_id              | FK device_models   | the real product (interfaces, protocols, default port)                        |
| serial                       | text               | hidden from Operator/Viewer by default                                        |
| location                     | text               |                                                                               |
| ip, port, mac                | inet, int, macaddr |                                                                               |
| link                         | enum               | ethernet   wifi   lte                                                         |
| field_protocol               | text               | Modbus RTU (RS-485), Modbus TCP, LoRaWAN                                      |
| uplink                       | enum               | mqtt_push   https_push   modbus_tcp_poll                                      |
| polling_s                    | integer            | 15, 30, 60                                                                    |
| serial_std                   | text null          | RS-485 2-wire   RS-485 4-wire   RS-232; null when no serial line              |
| baud                         | integer null       | 4800–115200; only for a serial field bus                                      |
| parity                       | enum null          | none   even   odd                                                             |
| data_bits, stop_bits         | smallint null      | 8/7, 1/2 — must match every meter on the line                                 |
| timeout_ms, retries          | integer            | Modbus reply timeout (default 1000) and retries (default 2)                   |
| time_sync                    | enum               | ntp   server                                                                  |
| firmware, firmware_available | text               | an update is offered when they differ                                         |
| installed_on                 | date               |                                                                               |
| status                       | enum               | online   weak   offline   maintenance (derived, see 6.6)                      |
| signal_pct                   | smallint null      |                                                                               |
| last_seen_at                 | timestampz null    | "4 s ago"                                                                     |

**device\_credentials** The device token — how a device proves who it is

| Column       | Type            | Notes                                                   |
|--------------|-----------------|---------------------------------------------------------|
| device_id    | FK PK           |                                                         |
| client_id    | text unique     | {orgId}.{deviceCode}, e.g. FEBINO.GW-07 (MQTT username) |
| token_hash   | char(64) unique | SHA-256 of the token; the token itself is never stored  |
| token_prefix | char(8)         | "edt_" + first 4 chars — to recognise a token in logs   |
| token_last4  | char(4)         | shown masked: edt_••••a91F                              |

| Column                 | Type                 | Notes                                  |
|------------------------|----------------------|----------------------------------------|
| created_at, created_by | timestampz, FK users | generated together with the device     |
| rotated_at, rotated_by | timestampz null      | Edit device → Regenerate token         |
| last_used_at           | timestampz null      |                                        |
| cert_fingerprint       | text null            | optional mutual TLS instead of a token |

### register\_maps Modbus register layout per meter model (for polled devices)

| Column              | Type                  | Notes                                                                                                                      |
|---------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------|
| id                  | uuid PK               |                                                                                                                            |
| name                | text unique           | Schneider PM5xxx, Schneider iEM3xxx, Secure Elite 4xx, DSE GenComm ...                                                     |
| registers           | jsonb                 | [(param:'kw', address:3060, fc:3, type:'float32', order:'ABCD', scale:1, unit:'kW'), ...] — 1-based as in the maker's list |
| source, verified_at | text, timestampz null | document the addresses came from; set when checked on a real meter                                                         |
| is_global           | boolean               | shared library; tenants may add their own                                                                                  |

### cost\_centres For allocating energy cost (Analytics, sub-meters)

| Column    | Type        | Notes                                                        |
|-----------|-------------|--------------------------------------------------------------|
| id        | uuid PK     |                                                              |
| tenant_id | FK          |                                                              |
| code      | varchar(16) | CC-PROD, CC-UTIL, CC-ADMIN, CC-IT, CC-FAC; unique per tenant |
| name      | text        | Production, Utilities ...                                    |

### meters Energy meters and sub-meters (one table, parent\_id makes a sub-meter)

| Column                           | Type                 | Notes                                                                                    |
|----------------------------------|----------------------|------------------------------------------------------------------------------------------|
| id                               | uuid PK              |                                                                                          |
| tenant_id, site_id               | FK                   |                                                                                          |
| code                             | varchar(20)          | MTR-1001 for meters, SM-1001-01 for sub-meters; unique per tenant                        |
| parent_id                        | FK meters null       | null = main meter (counts toward 'meters' limit); set = sub-meter ('subMeters' limit)    |
| name                             | text                 | Main Incomer, Feeder A · Production Line 1                                               |
| kind                             | enum                 | consumer   generator (solar, DG) — generators show 'Generating'                          |
| category                         | enum                 | incomer   hvac   lighting   process   utility   other                                    |
| cost_centre_id                   | FK cost_centres null | mainly for sub-meters                                                                    |
| meter_model_id                   | FK meter_models null | the real product (accuracy class, comms, supported wiring)                               |
| location                         | text                 |                                                                                          |
| meter_class                      | enum                 | basic   multi   pq   dc   dg — required; defaults from the model                         |
| energy_direction                 | enum                 | import (incoming feeder)   export (outgoing feeder) — required when the meter is created |
| serial_no                        | text null            | from the meter label                                                                     |
| wiring                           | enum                 | 3P4W   3P3W   1P2W — limited to the model's list                                         |
| pt_ratio                         | text null            | e.g. 11000/110; null for LT (415 V direct)                                               |
| commissioned_on, calibration_due | date null            | commissioning record                                                                     |
| phase                            | smallint             | 1 = single-phase 2-wire 230 V L-N; 3 = three-phase 4-wire 415 V L-L                      |
| nominal_v                        | numeric              | 230 or 415 (for formulas and limits)                                                     |
| ct_primary, ct_secondary         | integer null         | CT 400/5 → 400, 5; null = direct (e.g. Direct 63 A)                                      |
| direct_a                         | integer null         | rating for direct-connected sub-meters                                                   |
| rated_kw                         | numeric null         | for Load % = kW / rated                                                                  |
| device_id                        | FK devices null      | the device that reads it (may differ from the parent's device)                           |
| modbus_slave_id                  | smallint null        | 1–247; unique together with device_id                                                    |

| Column          | Type                  | Notes                                                                                                     |
|-----------------|-----------------------|-----------------------------------------------------------------------------------------------------------|
| register_map_id | FK register_maps null | defaults from model                                                                                       |
| register_offset | integer null          | multi-channel meters: channel's base register (sub-meter form)                                            |
| mode            | enum                  | active   maintenance   disabled — the only status a user sets                                             |
| icon, color     | text                  | UI only (i-cpu, --cyan)                                                                                   |
| status          | enum                  | online   idle   offline   maintenance   generating   standby — derived from readings + mode, never edited |
| archived_at     | timestamptz null      | keeps history when a meter is removed                                                                     |

### energy\_targets Monthly target & allocation (Analytics → Edit target & allocation)

| Column              | Type          | Notes                                        |
|---------------------|---------------|----------------------------------------------|
| id                  | uuid PK       |                                              |
| tenant_id, meter_id | FK            |                                              |
| month               | date          | first day of month; unique (meter_id, month) |
| target_kwh          | numeric(12,1) | > 0                                          |
| cost_centre_id      | FK null       |                                              |
| allocation_pct      | numeric(5,2)  | 0–100; share charged to the cost centre      |
| tariff_mode         | enum          | tod   flat                                   |
| notes               | text null     | reason for target change                     |
| set_by              | FK users      |                                              |

### virtual\_meter\_members Virtual meters (Σ nodes on the Connections canvas)

| Column     | Type                     | Notes                                          |
|------------|--------------------------|------------------------------------------------|
| virtual_id | FK meters (kind=virtual) | a virtual meter is a meters row without device |
| member_id  | FK meters                | a meter, sub-meter or another virtual meter    |
| sign       | smallint                 | +1 add, -1 subtract (e.g. Incomer – Solar)     |

### connection\_flows Saved and deployed versions of the Connections canvas

| Column                   | Type              | Notes                                                               |
|--------------------------|-------------------|---------------------------------------------------------------------|
| id                       | uuid PK           |                                                                     |
| tenant_id                | FK                |                                                                     |
| version                  | integer           | increments on each deploy                                           |
| nodes                    | jsonb             | [[id, type: submeter meter virtual device, name, ref, x, y, props]] |
| wires                    | jsonb             | [[from, to, kind: direct sub link]]                                 |
| is_draft                 | boolean           | one draft per tenant; deploy creates a new version                  |
| deployed_by, deployed_at | uuid, timestamptz |                                                                     |

## 1.4 Telemetry (time-series)

### readings TimescaleDB hypertable — one row per meter per reading

| Column          | Type        | Notes                                               |
|-----------------|-------------|-----------------------------------------------------|
| time            | timestamptz | meter/gateway timestamp (UTC); chunk interval 1 day |
| tenant_id       | varchar(12) | for RLS and per-tenant queries                      |
| meter_id        | uuid        | PK = (meter_id, time) — duplicates are ignored      |
| kw, kvar, kva   | real        | total active / reactive / apparent power            |
| pf              | real        | total power factor (–1...1)                         |
| v1, v2, v3      | real        | phase-to-neutral volts; 1φ meters fill only v1      |
| v12, v23, v31   | real null   | line-to-line volts (3φ only)                        |
| i1, i2, i3, i_n | real        | phase and neutral amps; 1φ meters fill only i1      |

| Column                                         | Type                  | Notes                                                                    |
|------------------------------------------------|-----------------------|--------------------------------------------------------------------------|
| hz                                             | real                  | frequency                                                                |
| thd_v, thd_i                                   | real                  | % total harmonic distortion (pq meters; per phase in thd_v1–3, thd_i1–3) |
| thd_v1, thd_v2, thd_v3, thd_i1, thd_i2, thd_i3 | real null             | per-phase THD for the Trends harmonics charts                            |
| fuel_pct, run_hours                            | real null             | DG set controllers only                                                  |
| kwh_imp, kwh_exp                               | double precision      | cumulative energy registers (import / export)                            |
| kvarh                                          | double precision null |                                                                          |
| quality                                        | smallint              | bit flags: 1 estimated, 2 out-of-range, 4 register reset, 8 late         |
| received_at                                    | timestamptz           | server time (for ingest-lag monitoring)                                  |

## 1.5 Tariffs, shifts and schedules

**tariffs** Time-of-Day tariff slots T1–T5 (Settings → Tariff slabs)

| Column               | Type                       | Notes                                                                            |
|----------------------|----------------------------|----------------------------------------------------------------------------------|
| id                   | uuid PK                    |                                                                                  |
| tenant_id, site_id   | FK (site null = all sites) |                                                                                  |
| slot                 | varchar(4)                 | T1 ... T5                                                                        |
| name                 | text                       | Off-peak (night), Morning peak, Evening peak ...                                 |
| start_time, end_time | time                       | 22:00 → 06:00 crosses midnight                                                   |
| days                 | smallint                   | bit mask Mon=1 ... Sun=64 (Mon – Sat = 63)                                       |
| energy_rate          | numeric(8,2)               | ₹ per kWh, e.g. 6.80, 9.20, 9.60                                                 |
| demand_charge        | numeric(8,2)               | ₹ per kVA of maximum demand per month                                            |
| valid_from, valid_to | date                       | keeps expired slabs (FY 2025-26) for history                                     |
| status               | derived                    | current if today is within valid_from...valid_to, otherwise expired (not stored) |

**tariff\_instances** Each run of a ToD slot (Energy cost → Instances)

| Column                    | Type         | Notes                                                         |
|---------------------------|--------------|---------------------------------------------------------------|
| id                        | uuid PK      |                                                               |
| tenant_id, site_id        | FK           |                                                               |
| code                      | varchar(20)  | TI-1008-T2; TI-1005-T1E = Monday's early T1 after Sunday flat |
| tariff_id                 | FK tariffs   | the slab in force when it ran (rate is copied below)          |
| starts_at, ends_at        | timestamptz  | T1 22:00 → 06:00 is one run, counted on the day it starts     |
| status                    | enum         | upcoming   running   closed                                   |
| rate                      | numeric(8,2) | ₹/kWh snapshot                                                |
| energy_kwh, cost, peak_kw | numeric null | updated live while running, final when closed                 |

**shifts** Shift definitions (Shift page)

| Column               | Type          | Notes                                                         |
|----------------------|---------------|---------------------------------------------------------------|
| id                   | uuid PK       |                                                               |
| tenant_id            | FK            |                                                               |
| code, name           | text          | A / Shift A                                                   |
| start_time, end_time | time          | end ≤ start means it ends the next day                        |
| break_min            | smallint      | 0   30   45   60                                              |
| count_on             | enum          | start_day   end_day — which day an overnight shift belongs to |
| supervisor_id        | FK users null |                                                               |
| grace_min            | smallint      | 0–60 late-arrival grace                                       |
| energy_target_kwh    | numeric null  | per shift run                                                 |

| Column | Type | Notes                                  |
|--------|------|----------------------------------------|
| status | enum | active   inactive                      |
| color  | text | amber   orange   violet   cyan   green |

### shift\_schedules Repeating plans (Schedule page)

| Column             | Type       | Notes                                                                 |
|--------------------|------------|-----------------------------------------------------------------------|
| id                 | uuid PK    |                                                                       |
| tenant_id          | FK         |                                                                       |
| name               | text       |                                                                       |
| pattern            | jsonb      | weekly or rotation: [{day/offset, shift_id}]                          |
| shift_id           | FK         | the form's Shift                                                      |
| days               | smallint   | Repeats on (bit mask) — the days a shift runs come from its schedules |
| pattern            | jsonb null | rotations later: [{offset, shift_id}]                                 |
| starts_on, ends_on | date       | Effective from / Until                                                |
| status             | enum       | active   paused                                                       |

### shift\_assignments Who works which shift (Schedule → Assign staff to shift)

| Column                       | Type    | Notes                                                   |
|------------------------------|---------|---------------------------------------------------------|
| id                           | uuid PK |                                                         |
| tenant_id, user_id, shift_id | FK      |                                                         |
| schedule_id                  | FK null |                                                         |
| from_date, to_date           | date    | to ≥ from; one user cannot have overlapping assignments |

### shift\_instances Each actual run of a shift (Instance page)

| Column                            | Type             | Notes                                                             |
|-----------------------------------|------------------|-------------------------------------------------------------------|
| id                                | uuid PK          |                                                                   |
| tenant_id, shift_id, schedule_id  | FK               |                                                                   |
| code                              | varchar(20)      | SI-1008-A                                                         |
| starts_at, ends_at                | timestamptz      | planned times, overnight resolved                                 |
| actual_start, actual_end          | timestamptz null | Planned / actual column                                           |
| staff_count                       | smallint         | from shift_assignments                                            |
| production_units                  | numeric null     | entered by supervisor; kWh / unit = energy_kwh / production_units |
| status                            | enum             | planned   running   closed                                        |
| energy_kwh, cost, peak_kw, avg_pf | numeric null     | snapshot written when the instance closes                         |

## 1.6 Alerts and notifications

### alert\_rules Per-meter rules (meter page → Alert rules; same shape the UI already saves)

| Column    | Type        | Notes                                                       |
|-----------|-------------|-------------------------------------------------------------|
| id        | uuid PK     |                                                             |
| tenant_id | FK          |                                                             |
| meter_id  | FK null     | null for tenant-wide rules (Alerts page)                    |
| code      | varchar(10) | AR-7F2                                                      |
| name      | text        | 1–80 chars                                                  |
| severity  | enum        | critical   warning   info                                   |
| enabled   | boolean     |                                                             |
| scope     | enum        | meter   subs   meter_and_subs   all   all_meters   all_subs |
| sub_ids   | uuid[]      | selected sub-meters ([] = all)                              |

| Column     | Type     | Notes                                                                                             |
|------------|----------|---------------------------------------------------------------------------------------------------|
| evaluate   | enum     | live   15m   1h   shift   day — live checks every reading; others check the window when it closes |
| cond       | jsonb    | {mode: guided expression, match: all any, rows:[{param, op, v1, v2}], expr, hold}                 |
| active     | jsonb    | {mode: always shifts slots custom, shifts, slots, from, to, days}                                 |
| notify     | jsonb    | {channels: app email sms whatsapp webhook, roles, users, emails, repeat, resolve, escalate}       |
| compiled   | jsonb    | server-validated form of cond (parsed expression, variable list)                                  |
| created_by | FK users |                                                                                                   |

**alert\_rule\_state** Where each rule stands for each meter it watches (also cached in Redis)

| Column                         | Type            | Notes                        |
|--------------------------------|-----------------|------------------------------|
| rule_id, meter_id              | PK              |                              |
| state                          | enum            | ok   pending   firing        |
| since                          | timestampz      | when the current state began |
| last_value                     | jsonb           | values that made it true     |
| alert_id                       | FK alerts null  | the open alert while firing  |
| last_notified_at, escalated_at | timestampz null |                              |

**alerts** The Alert log

| Column                                  | Type          | Notes                                                              |
|-----------------------------------------|---------------|--------------------------------------------------------------------|
| id                                      | uuid PK       |                                                                    |
| tenant_id                               | FK            |                                                                    |
| rule_id                                 | FK null       | null for system alerts (device offline, firmware, budget)          |
| meter_id, device_id                     | FK null       |                                                                    |
| title                                   | text          | THD above 8%                                                       |
| severity                                | enum          |                                                                    |
| category                                | enum          | threshold   power_quality   energy   device   communication   info |
| value_text, limit_text                  | text          | 9.1% / 8%                                                          |
| status                                  | enum          | open   acknowledged   resolved                                     |
| assigned_to                             | FK users null | Update alert → Assign to                                           |
| raised_at, acknowledged_at, resolved_at | timestampz    |                                                                    |
| acknowledged_by, resolved_by            | FK users null | null = auto-resolved                                               |
| note                                    | text null     | Resolution note                                                    |

**alert\_events** Timeline of an alert (raised, notified, escalated, acknowledged, resolved)

| Column   | Type       | Notes |
|----------|------------|-------|
| id       | bigserial  |       |
| alert_id | FK         |       |
| type     | text       |       |
| detail   | jsonb      |       |
| at       | timestampz |       |

**notifications** Every message sent, and the bell-icon inbox

| Column    | Type    | Notes                                     |
|-----------|---------|-------------------------------------------|
| id        | uuid PK |                                           |
| tenant_id | FK      |                                           |
| user_id   | FK null | null for plain email addresses / webhooks |

| Column           | Type             | Notes                                  |
|------------------|------------------|----------------------------------------|
| alert_id         | FK null          |                                        |
| channel          | enum             | app   email   sms   whatsapp   webhook |
| to               | text             |                                        |
| title, body      | text             |                                        |
| status           | enum             | queued   sent   failed                 |
| error            | text null        |                                        |
| sent_at, read_at | timestamptz null | read_at for in-app                     |

#### notification\_preferences Per user

| Column      | Type       | Notes                                   |
|-------------|------------|-----------------------------------------|
| user_id     | FK PK      |                                         |
| channels    | jsonb      | per severity, e.g. critical: [app, sms] |
| quiet_hours | jsonb null | except critical                         |

## 1.7 Audit, exports and device events

#### audit\_log Who changed what (append-only)

| Column               | Type         | Notes                                                      |
|----------------------|--------------|------------------------------------------------------------|
| id                   | bigserial    |                                                            |
| tenant_id            | varchar null | null for platform actions                                  |
| actor_type, actor_id | enum, uuid   | user   platform_admin   device   system                    |
| action               | text         | user.invite, meter.update, flow.deploy, tenant.suspend ... |
| entity, entity_id    | text         |                                                            |
| before, after        | jsonb        | changed fields only; secrets removed                       |
| ip                   | inet         |                                                            |
| at                   | timestamptz  |                                                            |

#### export\_jobs Large CSV / PDF / Excel exports run in the background

| Column             | Type        | Notes                                             |
|--------------------|-------------|---------------------------------------------------|
| id                 | uuid PK     |                                                   |
| tenant_id, user_id | FK          |                                                   |
| kind               | enum        | energy   alerts   readings   users   shift_energy |
| format             | enum        | csv   pdf   xlsx                                  |
| params             | jsonb       | {from, to, meters, columns}                       |
| status             | enum        | queued   running   done   failed                  |
| file_key           | text null   | object-storage key                                |
| expires_at         | timestamptz | files deleted after 7 days                        |

#### device\_messages Telemetry log — every message per meter and polling cycle (hypertable, kept 7 days)

| Column               | Type         | Notes                                    |
|----------------------|--------------|------------------------------------------|
| time                 | timestamptz  | read time (device clock)                 |
| tenant_id, device_id | FK           |                                          |
| meter_id             | FK null      | null when the slave is unknown           |
| slave                | smallint     | Modbus address                           |
| status               | enum         | ok   late   timeout   crc   out_of_range |
| latency_ms           | integer null | received_at – time; null for timeouts    |
| payload              | jsonb        | the reading exactly as sent (Raw button) |
| received_at          | timestamptz  |                                          |

**device\_commands** Commands sent to a device (Commands tab)

| Column                          | Type         | Notes                                                                              |
|---------------------------------|--------------|------------------------------------------------------------------------------------|
| id                              | uuid PK      |                                                                                    |
| code                            | varchar(12)  | CMD-3FT7                                                                           |
| tenant_id, device_id            | FK           |                                                                                    |
| command                         | enum         | ping   get_info   sync_time   set_polling   read_register   clear_buffer   restart |
| params                          | jsonb        | {slave, fc, address, count} / {interval}                                           |
| sent_by                         | FK users     | needs devices.command permission                                                   |
| status                          | enum         | queued   sent   success   failed   timeout (10 s without a reply)                  |
| reply                           | jsonb null   | the device's answer                                                                |
| reply_ms                        | integer null |                                                                                    |
| created_at, sent_at, replied_at | timestampz   |                                                                                    |

**publish\_destinations** Where a device's readings are forwarded (Data publish tab)

| Column               | Type              | Notes                                                             |
|----------------------|-------------------|-------------------------------------------------------------------|
| id                   | uuid PK           |                                                                   |
| tenant_id, device_id | FK                |                                                                   |
| name                 | text              | unique per device                                                 |
| type                 | enum              | mqtt   https                                                      |
| url                  | text              | mqtt://host:8883 or https://...                                   |
| topic                | text null         | MQTT topic template with {org}, {device}, {meter}; no + or #      |
| qos, retain          | smallint, boolean | MQTT only                                                         |
| method               | enum null         | POST   PUT (HTTPS)                                                |
| auth_secret          | text null         | encrypted; only a masked hint (Bearer •••• 7f2a) is ever returned |
| interval             | text              | reading   60   300   900 seconds                                  |
| format               | enum              | flat (one row per meter)   grouped (one message per device)       |
| fields               | text[]            | kw, kva, pf, v, i, hz, thd_v, thd_i, kwh                          |
| meter_ids            | uuid[]            | empty = all meters of the device                                  |
| enabled              | boolean           |                                                                   |

**publish\_attempts** Every publish (Data logs tab, together with device\_commands) — hypertable, kept 30 days

| Column             | Type       | Notes                                                                        |
|--------------------|------------|------------------------------------------------------------------------------|
| time               | timestampz |                                                                              |
| destination_id     | FK         |                                                                              |
| is_test            | boolean    | Test publish button                                                          |
| status             | enum       | success   retried (delivered after a retry)   failed (gave up after 3 tries) |
| detail             | text       | HTTP 200 / PUBACK · QoS 1 / HTTP 503 ...                                     |
| duration_ms, bytes | integer    |                                                                              |
| payload            | jsonb      | the message sent or attempted                                                |

**device\_events** Online/offline changes, firmware updates, reboots

| Column    | Type       | Notes                                                     |
|-----------|------------|-----------------------------------------------------------|
| id        | bigserial  |                                                           |
| device_id | FK         |                                                           |
| type      | text       | online   offline   weak_signal   firmware_update   config |
| detail    | jsonb      |                                                           |
| at        | timestampz |                                                           |

## 2. Modules

A module is one area of the application. Each row shows the pages it covers, the backend services behind it, the tables it uses and the permission key (module in the role permission matrix) that controls access.

| Module                      | Pages                                                                                                                                      | Services                                                                                                     | Tables                                                                                                                                             | Permission         |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Authentication & onboarding | login, register, verify-email, register-success, forgot-password, accept-invite, login-approval, awaiting-approval                         | AuthService, TenantService (register), InviteService                                                         | users, auth_sessions, otp_codes, invites, tenants                                                                                                  | — (public)         |
| Control Center (platform)   | admin-login, tenants, tenant, tenant-approvals                                                                                             | PlatformService, AuditService                                                                                | platform_admins, tenants, tenant_status_history, tenant_limits, audit_log                                                                          | platform realm     |
| Dashboard                   | dashboard                                                                                                                                  | AnalyticsService, LiveService + RealtimeGateway                                                              | readings_15m / 1h / 1d, meters, tariffs                                                                                                            | dashboard          |
| Energy meters               | meters, meter (tabs: Live, Energy, Trends, Sub-meters, Alerts, Device & connections)                                                       | MeterService, TelemetryService, LiveService, AlertRuleService, CatalogueService                              | meters, meter_models, register_maps, readings (+ aggregates), cost_centres, energy_targets, alert_rules                                            | meters             |
| Device management           | devices (incl. Telemetry logs for all devices), device (tabs: Overview, Meters, Telemetry, Commands, Data publish, Data logs, Connections) | DeviceService, CatalogueService, IngestionService, TelemetryLogService, DeviceCommandService, PublishService | devices, device_models, device_credentials, register_maps, device_messages, device_commands, publish_destinations, publish_attempts, device_events | devices            |
| Connections                 | connections                                                                                                                                | ConnectionService                                                                                            | connection_flows, meters, virtual_meter_members                                                                                                    | connections        |
| Analytics                   | analytics                                                                                                                                  | AnalyticsService, TariffService + CostService, ExportService                                                 | readings aggregates, energy_targets, cost_centres, export_jobs                                                                                     | analytics          |
| Alerts                      | alerts (and each meter's Alerts tab)                                                                                                       | AlertRuleService, AlertEngine, AlertService, NotificationService                                             | alert_rules, alert_rule_state, alerts, alert_events, notifications, notification_preferences                                                       | alerts             |
| Shift management            | shifts, shift-schedule, shift-instances                                                                                                    | ShiftService                                                                                                 | shifts, shift_schedules, shift_assignments, shift_instances                                                                                        | shifts             |
| Energy cost                 | tariffs, tariff-instances                                                                                                                  | TariffService + CostService, TariffInstanceService                                                           | tariffs, tariff_instances                                                                                                                          | settings (tariffs) |
| Users                       | users                                                                                                                                      | UserService + InviteService                                                                                  | users, user_sites, invites                                                                                                                         | users              |
| Roles & permissions         | roles                                                                                                                                      | RoleService + PermissionService                                                                              | roles, permissions, role_permissions, field_permissions                                                                                            | roles              |
| Settings                    | settings (Organisation, Sites, Notifications, Security, Plan & billing, Appearance)                                                        | TenantService, SiteService, AuthService (sessions), NotificationService (preferences)                        | tenants, sites, auth_sessions, notification_preferences                                                                                            | settings           |
| Cross-cutting               | every page                                                                                                                                 | AuditService, SchedulerService, ExportService, LiveService (WebSocket)                                       | audit_log, export_jobs                                                                                                                             | —                  |

## 3. Services

Each service is one backend module (NestJS) with a controller, business logic and a repository. Services marked worker run as separate processes so heavy traffic never slows page loads.

| Service                          | Purpose                                         | Main methods                                                                                                                |
|----------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| AuthService                      | Sign-in, sessions and one-time codes            | login, refresh, logout, me, requestOtp, verifyOtp, forgotPassword, resetPassword, switchTenant, listSessions, revokeSession |
| PlatformService (Control Center) | Tenant approvals, suspension and limits         | listTenants, getTenant, approve, reject, suspend, reactivate, setLimits, deleteTenant                                       |
| TenantService                    | Organisation profile, registration and limits   | register, getProfile, updateProfile, getLimits, usage, check                                                                |
| UserService + InviteService      | Team members                                    | list, get, invite, update, remove, resendInvite, acceptInvite, nextEmpId                                                    |
| RoleService + PermissionService  | Roles, module permissions and field permissions | listRoles, createRole, updateRole, deleteRole, getMatrix, setMatrix, getFieldPerms, setFieldPerms, can, canField            |
| SiteService                      | Locations                                       | list, create, update, remove                                                                                                |
| DeviceService                    | Gateways and loggers                            | list, get, create (→ device + token), update, remove, meters, regenerateToken, requestFirmwareUpdate                        |
| CatalogueService                 | Hardware catalogue                              | deviceModels, meterModels, registerMap, importRegisterMap, verifyRegisterMap                                                |
| MeterService                     | Meters, sub-meters and virtual meters           | tree, get, create, createSub, update, setMaintenance, archive, trends, energySummary                                        |
| ConnectionService                | Connections canvas                              | getFlow, saveDraft, validate, deploy, versions, rollback                                                                    |
| IngestionService (worker)        | Getting readings in                             | handleMqttMessage, handleHttpBatch, pollDevice, normalise, publish                                                          |
| TelemetryService                 | Storing and reading time-series                 | writeBatch, series, latest, energy, demand                                                                                  |
| LiveService + RealtimeGateway    | Realtime to the browser                         | onReading, snapshot, emit                                                                                                   |
| AnalyticsService                 | Reports and the Dashboard                       | dashboardSummary, consumption, subMeterReport, demand, powerQuality                                                         |
| TariffService + CostService      | Money                                           | list, create, update, remove, rateAt, cost                                                                                  |
| ShiftService                     | Shifts, schedules, instances                    | shifts CRUD, schedules CRUD, generateInstances, closeInstance, instanceEnergy                                               |
| AlertRuleService                 | Rule management                                 | list, create, update, remove, compile, test                                                                                 |
| AlertEngine (worker)             | Evaluating rules 24×7                           | onReading, onWindowClose, evaluate, transition                                                                              |
| AlertService                     | The Alert log                                   | list, acknowledge, acknowledgeAll, resolve, remove                                                                          |
| NotificationService (worker)     | Delivery                                        | enqueue, send, markRead, inbox                                                                                              |
| ExportService (worker)           | CSV / PDF / Excel                               | create, status, download                                                                                                    |
| SchedulerService                 | Timed jobs (BullMQ repeatable)                  | register jobs                                                                                                               |
| AuditService                     | Traceability                                    | record, list                                                                                                                |
| DeviceCommandService             | Commands to a device                            | send, list, get, onReply, timeout                                                                                           |
| PublishService (worker)          | Forwarding readings to other systems            | createDestination, updateDestination, removeDestination, test, preview, runInterval                                         |
| TelemetryLogService              | Device telemetry log                            | record, list, listAll, chartSeries, dataLogs                                                                                |
| TariffInstanceService            | Runs of Time-of-Day slots                       | generate, onReading, close, list, summary, get                                                                              |

**AuthService** — Sign-in, sessions and one-time codes

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"> <li>• Sign in with Org ID + username / email / mobile + password; checks tenant status (pending → awaiting approval page, suspended → reason and date) and user status.</li> <li>• Optional 2FA: password then 6-digit OTP; OTP-only sign-in by mobile/email.</li> <li>• Issues a 15-minute access JWT and a rotating refresh token in an httpOnly cookie; remember-me 30 days, otherwise 12 hours.</li> <li>• Forgot / reset password, verify email, login approval, switch tenant (same email in another active tenant), list and revoke sessions.</li> </ul> |
| <b>Main methods</b>     | login, refresh, logout, me, requestOtp, verifyOtp, forgotPassword, resetPassword, switchTenant, listSessions, revokeSession                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

**PlatformService (Control Center)** — Tenant approvals, suspension and limits

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"> <li>• Separate realm and 8-hour session for platform admins, MFA required.</li> <li>• Approve or reject pending registrations (emails the tenant admin), suspend with reason and optional end date, reactivate, delete.</li> <li>• Revokes every session of a suspended tenant immediately; scheduler lifts timed suspensions.</li> <li>• Edit limits per tenant with a reason; view a tenant's users and usage.</li> </ul> |
| <b>Main methods</b>     | listTenants, getTenant, approve, reject, suspend, reactivate, setLimits, deleteTenant                                                                                                                                                                                                                                                                                                                                                                          |

**TenantService** — Organisation profile, registration and limits

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"> <li>• Self-registration creates a pending tenant + its admin user (generates an 8-character Org ID or reserves the chosen slug).</li> <li>• Profile settings (name, GSTIN, address, utility, contract demand, time zone, CO2 factor).</li> <li>• LimitService.check(tenant, key) is called inside the same transaction as every create — the server, not the browser, enforces limits (409 LIMIT_REACHED).</li> </ul> |
| <b>Main methods</b>     | register, getProfile, updateProfile, getLimits, usage, check                                                                                                                                                                                                                                                                                                                                                                                             |

**UserService + InviteService** — Team members

|                         |                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"> <li>• List with search, role, status, department filters and server-side paging.</li> <li>• Invite (creates user in 'invited' state, emails a link), resend, accept (sets password), suspend, deactivate, delete.</li> <li>• Next employee ID, email unique per tenant, username unique platform-wide, site restrictions.</li> </ul> |
| <b>Main methods</b>     | list, get, invite, update, remove, resendInvite, acceptInvite, nextEmpId                                                                                                                                                                                                                                                                                                |

**RoleService + PermissionService** — Roles, module permissions and field permissions

|                         |                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"> <li>• CRUD custom roles; built-in roles are seeded per tenant and cannot be deleted.</li> <li>• Permission matrix per module/action; field permissions per role (view/edit).</li> <li>• A guard checks module permissions on every route; a response interceptor removes fields the role may not view and rejects writes to fields it may not edit.</li> </ul> |
| <b>Main methods</b>     | listRoles, createRole, updateRole, deleteRole, getMatrix, setMatrix, getFieldPerms, setFieldPerms, can, canField                                                                                                                                                                                                                                                                                  |

**SiteService** — Locations

|                         |                                                                                |
|-------------------------|--------------------------------------------------------------------------------|
| <b>Responsibilities</b> | • CRUD sites with utility, consumer number and contract demand; limit 'sites'. |
| <b>Main methods</b>     | list, create, update, remove                                                   |

**DeviceService** — Gateways and loggers

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"> <li>• CRUD devices; generates the device token on create and returns it once; regenerate token (old one revoked, device disconnected).</li> <li>• Lists the meters a device reads (with slave IDs); firmware info and update requests; status history.</li> <li>• Stores the serial line (standard, baud, parity, data / stop bits) and Modbus timeout / retries; checks the uplink is one the model supports.</li> <li>• Hidden fields (IP, MAC, serial) filtered by field permissions.</li> </ul> |
| <b>Main methods</b>     | list, get, create (→ device + token), update, remove, meters, regenerateToken, requestFirmwareUpdate                                                                                                                                                                                                                                                                                                                                                                                                                                   |

**CatalogueService** — Hardware catalogue

|                         |                                                                                                                                                                                                                                                                                                                                |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"> <li>• Lists real device models and meter models with their published specifications; tenants may add their own entries.</li> <li>• Register maps per meter model (1-based addresses, function code, data type, word order, unit); import from the maker's CSV; mark as verified.</li> </ul> |
| <b>Main methods</b>     | deviceModels, meterModels, registerMap, importRegisterMap, verifyRegisterMap                                                                                                                                                                                                                                                   |

**MeterService** — Meters, sub-meters and virtual meters

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"> <li>• Meter tree (main meters with their sub-meters), filters by site, status, consumer/generator.</li> <li>• Create meter / sub-meter with Supply (1φ / 3φ), CT ratio, rated kW, device and slave ID; limits 'meters' and 'subMeters'.</li> <li>• Validates that (device, slave ID) is free and that 1φ meters have no v2/v3 parameters in their rules.</li> <li>• Requires meter class and Incoming / Outgoing on create; wiring limited to the model's list; keeps the commissioning record (serial no., CT / PT, dates).</li> <li>• Trends series (kW, kVA, PF, per-phase THD) from readings_15m; import / export energy summary.</li> <li>• Maintenance mode on/off; archive instead of hard delete.</li> </ul> |
| <b>Main methods</b>     | tree, get, create, createSub, update, setMaintenance, archive, trends, energySummary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

**ConnectionService** — Connections canvas

|                         |                                                                                                         |
|-------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | • Load registry-based default flow, save draft, validate, deploy (versioned), list versions, roll back. |
| <b>Main methods</b>     | getFlow, saveDraft, validate, deploy, versions, rollback                                                |

**IngestionService (worker)** — Getting readings in

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"> <li>• MQTT consumer, HTTPS ingest endpoint and Modbus TCP poller all produce the same normalised reading.</li> <li>• Authenticates the device, maps slave ID → meter via a cached device map, applies scaling, rejects impossible values, flags register resets.</li> <li>• Deduplicates on (meter, time), stamps received_at, appends to Redis Stream 'readings', updates device last_seen.</li> </ul> |
| <b>Main methods</b>     | handleMqttMessage, handleHttpBatch, pollDevice, normalise, publish                                                                                                                                                                                                                                                                                                                                                                         |

**TelemetryService** — Storing and reading time-series

|                         |                                                                                                                                                                                                                                                                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"> <li>• DB writer consumer: batches stream entries and inserts with COPY (every 1 s or 500 rows).</li> <li>• Query API for charts: picks raw or the right aggregate, fills gaps, returns only requested parameters.</li> <li>• Daily jobs: shift instance snapshots, daily energy totals.</li> </ul> |
| <b>Main methods</b>     | writeBatch, series, latest, energy, demand                                                                                                                                                                                                                                                                                            |

**LiveService + RealtimeGateway** — Realtime to the browser

|                         |                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"> <li>• Consumes the stream, updates live:{meter} in Redis, computes load %, imbalance, <math>\Sigma</math> sub-meters and <math>\Delta</math> unmetered, today's kWh and cost.</li> <li>• Emits to Socket.IO rooms, at most once per second per meter, filtered by field permissions (e.g. no cost for Operators).</li> </ul> |
| <b>Main methods</b>     | onReading, snapshot, emit                                                                                                                                                                                                                                                                                                                                       |

**AnalyticsService** — Reports and the Dashboard

|                         |                                                                                                                                                                                                                                                                     |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"> <li>• Dashboard summary (live load, today's kWh, cost, CO2, max demand vs contract, peak time).</li> <li>• Consumption by meter / sub-meter / site / shift / day, sub-meter report, power quality, solar vs forecast.</li> </ul> |
| <b>Main methods</b>     | dashboardSummary, consumption, subMeterReport, demand, powerQuality                                                                                                                                                                                                 |

**TariffService + CostService** — Money

|                         |                                                                                                              |
|-------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | • CRUD ToD slots; splits energy by slot (in the site's time zone) and applies energy rate and demand charge. |
| <b>Main methods</b>     | list, create, update, remove, rateAt, cost                                                                   |

**ShiftService** — Shifts, schedules, instances

|                         |                                                                                                                                                                                                     |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | • CRUD shifts and schedules; generates shift instances 14 days ahead on the days each schedule repeats; resolves overnight shifts and the 'count on' day; closes instances and stores their energy. |
| <b>Main methods</b>     | shifts CRUD, schedules CRUD, generateInstances, closeInstance, instanceEnergy                                                                                                                       |

**AlertRuleService** — Rule management

|                         |                                                                                                                                                                                                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"> <li>• CRUD rules with the UI's JSON shape; validates parameters against the meter's phase; compiles expressions safely; limit 'alertRules'.</li> <li>• Dry-run: replays the last 24 hours and reports when the rule would have fired.</li> </ul> |
| <b>Main methods</b>     | list, create, update, remove, compile, test                                                                                                                                                                                                                                         |

**AlertEngine (worker)** — Evaluating rules 24×7

|                         |                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"> <li>• Live rules on every reading, window rules when a window closes, 'no readings' rules from the offline checker.</li> <li>• State machine with hold time, active windows, repeat, auto-resolve and escalation (section 7). Writes alerts and alert_events, queues notifications.</li> </ul> |
| <b>Main methods</b>     | onReading, onWindowClose, evaluate, transition                                                                                                                                                                                                                                                                                    |

**AlertService** — The Alert log

|                         |                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | • List/filter/page alerts, acknowledge, acknowledge all, resolve with note, delete (admin). |
| <b>Main methods</b>     | list, acknowledge, acknowledgeAll, resolve, remove                                          |

**NotificationService (worker)** — Delivery

|                         |                                                                                                                                                                                  |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | • Resolves roles → users, applies preferences and quiet hours, sends via app (WebSocket + inbox), email, SMS, WhatsApp or signed webhook; retries with back-off; records status. |
| <b>Main methods</b>     | enqueue, send, markRead, inbox                                                                                                                                                   |

**ExportService (worker)** — CSV / PDF / Excel

|                         |                                                                                                                                                                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | • Small exports are generated on request; large ranges (over 31 days or 50k rows) become a background job with a download link. Columns match the current exports (Date, Meter ID, Meter, Parent meter, Location, Phase, Energy, Peak, Cost, CO2). |
| <b>Main methods</b>     | create, status, download                                                                                                                                                                                                                           |

**SchedulerService** — Timed jobs (BullMQ repeatable)

|                         |                                                                                                                                                                                                                           |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | • Every 30 s: device and meter offline check. Every minute: window-rule evaluation, timed-suspension lift, escalations. Hourly: invite/OTP clean-up. Nightly: shift instance generation, export clean-up, usage counters. |
| <b>Main methods</b>     | register jobs                                                                                                                                                                                                             |

**AuditService** — Traceability

|                         |                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | • Interceptor records every create/update/delete with before/after (secrets removed); queryable by Tenant Admin and Auditor. |
| <b>Main methods</b>     | record, list                                                                                                                 |

**DeviceCommandService** — Commands to a device

|                         |                                                                                                                                                                                                                                                                                                                         |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"> <li>• Send ping, get info, sync time, set polling, read Modbus registers (read only), clear stored readings, restart.</li> <li>• Queued → Sent → Success / Failed / Timed out (10 s); replies stored with the command; restart / clear need confirmation and are audited.</li> </ul> |
| <b>Main methods</b>     | send, list, get, onReply, timeout                                                                                                                                                                                                                                                                                       |

**PublishService (worker)** — Forwarding readings to other systems

|                         |                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"> <li>• CRUD publish destinations per device (MQTT broker or HTTPS webhook, topic/URL, interval, payload format, fields, meters, on/off).</li> <li>• Builds each payload from the live cache at its interval, retries twice, records every attempt; Test publish; payload preview; 'Publish failing' alert after 3 failures.</li> </ul> |
| <b>Main methods</b>     | createDestination, updateDestination, removeDestination, test, preview, runInterval                                                                                                                                                                                                                                                                                      |

**TelemetryLogService** — Device telemetry log

|                         |                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"> <li>• Writes every message per meter and polling cycle with status (ok, late, timeout, crc, out of range), latency and raw payload; kept 7 days. Values follow the meter class.</li> <li>• Serves the device's Telemetry tab (log + chart), Device Management → Telemetry logs (all devices) and the Data logs tab.</li> </ul> |
| <b>Main methods</b>     | record, list, listAll, chartSeries, dataLogs                                                                                                                                                                                                                                                                                                                      |

**TariffInstanceService** — Runs of Time-of-Day slots

|                         |                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b> | <ul style="list-style-type: none"><li>• Creates the next 2 days of tariff instances from the slabs (Mon–Sat T1–T5, Sunday Flat; T1 22:00 → 06:00 counted on its start day).</li><li>• Adds live energy to the running instance, closes it from 15-minute aggregates; totals and By-slot summary for Energy cost → Instances.</li></ul> |
| <b>Main methods</b>     | generate, onReading, close, list, summary, get                                                                                                                                                                                                                                                                                         |

## 4. APIs

Base URL `/api/v1`, JSON, Authorization: Bearer <access token> (refresh token in an httpOnly cookie). The tenant always comes from the token. Paging: `?page=&size=` → { data, page, size, total }. Errors: { error: { code, message, field? } } — VALIDATION, UNAUTHENTICATED, FORBIDDEN, NOT\_FOUND, CONFLICT, LIMIT\_REACHED, RATE\_LIMITED. Realtime updates use the Socket.IO namespace `/live` (events: reading, meter.status, device.status, dashboard, alert, notification, session.revoked).

### 4.1 Auth

| Method       | Path                                                                   | Purpose                                                          |
|--------------|------------------------------------------------------------------------|------------------------------------------------------------------|
| POST         | <code>/auth/register</code>                                            | Register a tenant + admin (→ pending)                            |
| POST         | <code>/auth/verify-email</code>                                        | Confirm the 6-digit code                                         |
| POST         | <code>/auth/login</code>                                               | {orgId, login, password, remember} → token, or {challenge:'otp'} |
| POST         | <code>/auth/otp/request</code> · <code>/auth/otp/verify</code>         | OTP sign-in / 2FA step                                           |
| POST         | <code>/auth/refresh</code>                                             | Rotate tokens (cookie)                                           |
| POST         | <code>/auth/logout</code>                                              | Revoke this session                                              |
| POST         | <code>/auth/forgot-password</code> · <code>/auth/reset-password</code> | Reset flow                                                       |
| GET          | <code>/auth/me</code>                                                  | User, tenant, role permissions, field permissions, limits        |
| GET          | <code>/auth/memberships</code>                                         | Other tenants with the same email                                |
| POST         | <code>/auth/switch-tenant</code>                                       | {orgId}                                                          |
| GET · DELETE | <code>/auth/sessions/:id</code>                                        | Active sessions / sign out a device                              |
| GET · POST   | <code>/invites/:token[/accept]</code>                                  | Show and accept an invitation                                    |

### 4.2 Control Center (platform realm)

| Method | Path                                                                                                              | Purpose                         |
|--------|-------------------------------------------------------------------------------------------------------------------|---------------------------------|
| POST   | <code>/admin/auth/login</code>                                                                                    | Platform admin sign-in (+ MFA)  |
| GET    | <code>/admin/tenants?status=</code>                                                                               | Tenants and approvals           |
| GET    | <code>/admin/tenants/:id</code>                                                                                   | Profile, usage, limits, history |
| POST   | <code>/admin/tenants/:id/approve</code> · <code>/reject</code> · <code>/suspend</code> · <code>/reactivate</code> | {reason, until}                 |
| PUT    | <code>/admin/tenants/:id/limits</code>                                                                            | {users: 60, devices: 0, reason} |
| GET    | <code>/admin/tenants/:id/users</code>                                                                             | Read-only list                  |
| DELETE | <code>/admin/tenants/:id</code>                                                                                   | Soft delete                     |

### 4.3 Tenant, users, roles, sites

| Method               | Path                                                                                   | Purpose                           |
|----------------------|----------------------------------------------------------------------------------------|-----------------------------------|
| GET · PATCH          | <code>/tenant</code>                                                                   | Organisation profile              |
| GET                  | <code>/tenant/limits</code>                                                            | Limit, used and remaining per key |
| GET · POST           | <code>/users</code>                                                                    | List (filters, paging) / invite   |
| GET · PATCH · DELETE | <code>/users/:id</code>                                                                |                                   |
| POST                 | <code>/users/:id/resend-invite</code> · <code>/suspend</code> · <code>/activate</code> |                                   |
| GET · POST           | <code>/roles</code>                                                                    |                                   |
| PATCH · DELETE       | <code>/roles/:id</code>                                                                |                                   |
| GET · PUT            | <code>/roles/:id/permissions</code>                                                    | Module × action matrix            |
| GET · PUT            | <code>/roles/:id/field-permissions</code>                                              | module.field → {view, edit}       |

| Method                      | Path               | Purpose                      |
|-----------------------------|--------------------|------------------------------|
| GET · POST · PATCH · DELETE | /sites[:id]        |                              |
| GET · POST                  | /permissions       | Permission key catalogue     |
| GET · POST · PATCH · DELETE | /cost-centres[:id] |                              |
| PATCH · POST                | /me · /me/password | My profile / Change password |

## 4.4 Devices, meters, connections

| Method               | Path                                                                                  | Purpose                                                       |
|----------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------|
| GET · POST           | /devices                                                                              | POST returns { device, token } — the token only this once     |
| GET · PATCH · DELETE | /devices/:id                                                                          |                                                               |
| GET                  | /devices/:id/meters                                                                   | Meters it reads, with slave IDs                               |
| POST                 | /devices/:id/token/regenerate                                                         | New device token (shown once); old token revoked              |
| GET                  | /devices/:id/telemetry?from&to&slave&status&page&size                                 | Telemetry log (+ CSV with Accept: text/csv)                   |
| GET                  | /telemetry?device&meter&status&from&to&page&size                                      | Telemetry logs for all devices (Device Management)            |
| GET                  | /catalogue/device-models · /catalogue/meter-models                                    | Hardware catalogue for the Make & model lists                 |
| GET · POST           | /register-maps · /register-maps/:id                                                   | Register maps; POST imports a maker's list (CSV)              |
| GET · POST           | /devices/:id/commands                                                                 | Command history / send a command                              |
| GET                  | /devices/:id/commands/:cmdId                                                          | Status and reply                                              |
| GET · POST           | /devices/:id/publish-destinations                                                     |                                                               |
| PATCH · DELETE       | /devices/:id/publish-destinations/:destId                                             | Edit, enable / pause, delete                                  |
| POST                 | /devices/:id/publish-destinations/:destId/test                                        | Test publish now                                              |
| GET                  | /devices/:id/publish-destinations/:destId/preview                                     | Next payload                                                  |
| GET                  | /devices/:id/data-logs?from&to&kind&status&q&page&size                                | Publishes + command replies                                   |
| GET                  | /devices/:id/events                                                                   | Status history                                                |
| GET                  | /meters?view=tree&site=&status=&kind=                                                 | Meter tree (replaces ED_METERS)                               |
| POST                 | /meters                                                                               | Create main meter                                             |
| POST                 | /meters/:id/sub-meters                                                                | Create sub-meter                                              |
| GET · PATCH · DELETE | /meters/:id                                                                           | Delete = archive                                              |
| GET                  | /meters/:id/live                                                                      | Latest values from the live cache                             |
| GET                  | /meters/:id/readings?from&to&interval=auto 1m 15m 1h 1d&params=kw,pf                  | Chart series                                                  |
| GET                  | /meters/:id/summary?period=today month                                                | kWh, cost, peak, PF, $\Sigma/\Delta$ , import / export energy |
| GET                  | /meters/:id/trends?from&to&params=kw,kva,pf,thd_v1,thd_v2,thd_v3,thd_i1,thd_i2,thd_i3 | Trends tab charts (15-minute)                                 |
| GET · PUT            | /connections/flow                                                                     | Current draft                                                 |
| POST                 | /connections/flow/validate · /deploy                                                  |                                                               |
| GET                  | /connections/flow/versions                                                            |                                                               |
| POST                 | /connections/flow/versions/:v/rollback                                                |                                                               |

## 4.5 Analytics, tariffs, shifts

| Method                      | Path                                                        | Purpose                           |
|-----------------------------|-------------------------------------------------------------|-----------------------------------|
| GET                         | /dashboard/summary                                          | Tiles on the Dashboard            |
| GET                         | /analytics/consumption?from&to&groupBy=meter site shift day |                                   |
| GET                         | /analytics/sub-meter-report?from&to                         | Sub-meter consumption report      |
| GET                         | /analytics/demand?month=                                    | 15-min maximum demand vs contract |
| GET                         | /analytics/power-quality?from&to                            | PF, THD, voltage, imbalance       |
| GET · PUT                   | /energy-targets/:meterId/:month                             | Target & allocation               |
| GET · POST · PATCH · DELETE | /tariffs[:id]                                               | ToD slots                         |
| GET                         | /tariffs/active?at=                                         | Slot and rate now                 |
| GET                         | /tariff-instances?from&to&slot&status&page&size             | Tariff instances list             |
| GET                         | /tariff-instances/summary?from&to                           | Totals and By-slot table          |
| GET                         | /tariff-instances/:id                                       | Detail with per-meter split       |
| GET · POST · PATCH · DELETE | /shifts[:id] · /shift-schedules[:id]                        |                                   |
| GET · POST · DELETE         | /shift-assignments                                          | Assign staff                      |
| GET                         | /shift-instances?from&to                                    |                                   |
| GET                         | /shift-instances/:id/energy                                 | Energy, cost, peak per meter      |

## 4.6 Alerts, notifications, exports, audit

| Method             | Path                                              | Purpose                          |
|--------------------|---------------------------------------------------|----------------------------------|
| GET                | /alerts?status&severity&category&search&page&size | Alert log                        |
| POST               | /alerts/:id/acknowledge · /resolve                | {note}                           |
| POST               | /alerts/acknowledge-all                           |                                  |
| DELETE             | /alerts/:id                                       | Tenant Admin only                |
| GET · POST         | /alert-rules?meter=                               |                                  |
| GET · PUT · DELETE | /alert-rules/:id                                  |                                  |
| POST               | /alert-rules/:id/test                             | 24-hour dry run                  |
| GET                | /notifications?unread=true                        | Bell inbox                       |
| POST               | /notifications/:id/read · /read-all               |                                  |
| GET · PUT          | /me/notification-preferences                      |                                  |
| POST               | /exports                                          | {kind, format, from, to, meters} |
| GET                | /exports/:id · /exports/:id/download              |                                  |
| GET                | /audit?entity&actor&from&to                       | Tenant Admin, Auditor            |

## 4.7 Device ingestion (device credentials, not user tokens)

| Method | Path                                     | Purpose                                                     |
|--------|------------------------------------------|-------------------------------------------------------------|
| POST   | /ingest/v1/readings                      | Batch of readings (HTTPS push gateways)                     |
| POST   | /ingest/v1/heartbeat                     | Status, signal, firmware                                    |
| POST   | /internal/mqtt/auth · /internal/mqtt/acl | Called by EMQX to authorise devices (internal network only) |